Components of the Hadoop Architecture

In this train, we'll dive into the key components of the Hadoop architecture, including HDFS, MapReduce, and YARN, and understand their roles in Big Data processing.

Learning Objectives

By the end of this train, you should be able to:

- Identify and describe the various components of the Hadoop architecture.

Hadoop Distributed File System (HDFS)

HDFS is the storage layer of Hadoop. It's designed to store very large files across a cluster of commodity hardware.

It divides files into blocks and distributes them across multiple nodes, ensuring fault tolerance through data replication.

The key components of HDFS are:

- **NameNode:** This is the master node that manages the file system namespace and metadata.
  Note: **The NameNode does not store data itself—it only tracks where data blocks are stored across the cluster.**
- **DataNode:** The worker nodes that store the actual data blocks.

  They handle read and write requests from clients and report back to the NameNode.

Yet Another Resource Negotiator (YARN)

YARN is the resource management layer. It's responsible for managing cluster resources and scheduling applications.

YARN supports multiple data processing engines—such as MapReduce, Spark, and Tez—on the same Hadoop cluster, providing flexibility and improved resource utilization.

The key components of YARN include:

- **ResourceManager**: The master component that manages the allocation of cluster resources.
- **NodeManager**: These are the worker components that manage resources on individual nodes.
- **ApplicationMaster**: Manages the applications that run on the YARN framework.

MapReduce

MapReduce is a programming model and framework for processing large datasets in parallel.

It works by dividing data processing into two main phases:

- **Map** : This phase processes input data and generates key-value pairs.
- **Reduce** : This phase aggregates and combines the output from the Map phase to produce the final result.

MapReduce enables distributed data processing and greatly speeds up the transformation of large datasets.

Conclusion

These components we've covered work together to enable the efficient and scalable processing of big data.

▼Key Takeaways ✖

- **Takeaway 1:** HDFS handles the storage of massive datasets.
- **Takeaway 2:** YARN manages the cluster resources.
- **Takeaway 3:** MapReduce provides the framework for parallel data processing.